

Computer Science & Information Technology

Discrete Mathematics

DPP: 1

Mathematical Logic

Q1 Which of the following is tautology?

- (A) $(\sim p \wedge (p \rightarrow q)) \rightarrow \sim q$
 (B) $\sim (p \rightarrow q) \rightarrow \sim q$
 (C) $[(\sim p \wedge q) \wedge (q \rightarrow (p \rightarrow q))] \rightarrow \sim r$
 (D) None of these

Q2 The statement $[P \vee (P \leftrightarrow Q) \vee Q]$ is equivalent to

- (A) P (B) Q
 (C) A tautology (D) $(P \wedge Q)$

Q3 Consider the following statement

$$S_1: [(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (r \Rightarrow p)$$

$$S_2: [(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$$

Which of the following is/are correct?

- (A) S_1 is contingency
 (B) S_2 is tautology
 (C) S_1 and S_2 both contingency
 (D) S_1 and S_2 both Tautology

Q4 Which of the following is valid?

$$S_1: p \Rightarrow (q \vee r) \equiv (p \Rightarrow q) \vee (p \Rightarrow r)$$

$$S_2: p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)$$

- (A) S_1 is valid and S_2 is not valid
 (B) S_1 is not valid and S_2 is valid
 (C) Both S_1 and S_2 are valid
 (D) Neither S_1 and S_2 is valid

Q5 Which of the following is/are true.

- (A) $\{R \rightarrow S, P \rightarrow Q, R \vee P\} \Rightarrow S \vee Q$
 (B) $\{\sim R \rightarrow (S \rightarrow \sim T), \sim R \vee W, \sim P \rightarrow S, \sim W\} \Rightarrow T \rightarrow P$
 (C) $\{\sim P \wedge Q, Q \rightarrow (P \rightarrow R)\} \Rightarrow \sim R$
 (D) $\{P \rightarrow R, P, Q \vee \sim R\} \Rightarrow Q$

Q6 Consider the following logical inferences.

I_1 : "I will study discrete math or I will study English literature"

"I will not study discrete math"

Inference: I will not study English literature

I_2 : "If A works hard then B or C will enjoy

themselves"

"If B enjoys himself then A will not work hard"

"If C enjoys himself then D will not enjoy himself"

Inference: - If A work hard then D will not enjoy himself

- (A) Both I_1 and I_2 are correct inferences
 (B) I_1 is correct but I_2 is not a correct inference
 (C) I_1 is not correct but I_2 is a correct inference
 (D) Both I_1 and I_2 are not correct inferences

Q7 Which of the following is/are logical equivalence?

I. $\sim (p \rightarrow q)$

II. $\sim (p \rightarrow q) \wedge (q \rightarrow r)$

III. $p \wedge \sim q$

IV. $(p \vee q) \rightarrow r$

- (A) I and II (B) I and III
 (C) II and IV (D) II and III

Q8 Consider the following statement

$$S_1: \sim (p \leftrightarrow q)$$

$$S_2: p \leftrightarrow \sim q$$

Which of the following is correct?

- (A) S_1 is tautology
 (B) S_2 is contradiction
 (C) S_1 is equivalent to S_2
 (D) None of these

Q9 Consider the following statement

$$S_1: \sim (p \vee (\sim p \wedge q))$$

$$S_2: \sim p \wedge \sim q$$

Which of the following is correct?

- (A) S_1 is tautology
 (B) S_2 is contradiction
 (C) S_1 is equivalence to S_2
 (D) S_1 is not equivalence to S_2

Q10 $(p \rightarrow (q \rightarrow r)) \leftrightarrow ((p \rightarrow q) \rightarrow r)$ is

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(A) Tautology
(B) Contingency

(C) Contradiction
(D) Can't be determined.



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Answer Key

Q1 (B)

Q2 (C)

Q3 (A, B)

Q4 (C)

Q5 (A, B, D)

Q6 (C)

Q7 (A, B, D)

Q8 (C)

Q9 (C)

Q10 (B)



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Hints & Solutions

Q1 Text Solution:

- (a) $(\sim p \wedge (p \rightarrow q)) \rightarrow q$
 $\equiv (\sim p \wedge (\sim p \vee q)) \rightarrow \sim q$
 $\equiv ((\sim p \wedge \sim p) \vee (\sim p \wedge q)) \rightarrow \sim q$
 $\equiv \sim p \rightarrow \sim q$
 $\equiv p \vee \sim q$ (Not a tautology)
- (b) $(\sim(p \rightarrow q)) \rightarrow \sim q$
 $\equiv (\sim(\sim p \vee q)) \rightarrow \sim q$
 $\equiv (p \wedge \sim q) \rightarrow \sim q$
 $\equiv \sim(p \wedge \sim q) \vee \sim q$
 $\equiv \sim p \vee q \vee \sim q$
 $\equiv T$ (tautology)
- (c) $[(\sim p \wedge q) \wedge (q \rightarrow (p \rightarrow q))] \rightarrow \sim r$
 $\equiv [(\sim p \wedge q) \wedge (\sim q \vee \sim p \vee q)] \rightarrow \sim r$
 $\equiv (\sim p \wedge q) \rightarrow \sim r$
 $\equiv p \vee \sim q \vee \sim r$ (Not tautology)

Q2 Text Solution:

$$p \vee (p \leftrightarrow q) \vee Q$$

$$\equiv p \vee (p \wedge Q) \vee (\sim p \wedge \sim Q) \vee Q$$

$$\equiv p \vee (\sim p \wedge \sim q) \vee Q$$

$$\equiv T$$

Q3 Text Solution:

$$S_1: [(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (r \rightarrow p)$$

$$\equiv [(\sim p \vee q) \wedge (\sim q \vee r)] \rightarrow (\sim r \vee p)$$

$$\equiv [(\sim p \wedge \sim q) \vee (\sim p \wedge r) \vee (q \wedge r)] \rightarrow (\sim r \vee p)$$

$$\equiv [(p \vee q) \wedge (p \vee \sim r) \wedge (\sim q \vee \sim r)] \vee (\sim r \vee p)$$

$$\equiv p \vee \sim r$$
 (Contingency)

$$S_2: [(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

It is the valid transitivity property of implication.

Q4 Text Solution:

$$S_1: p \rightarrow (q \vee r) \equiv [(p \rightarrow q) \vee (p \rightarrow r)]$$

$$\text{LHS: } p \rightarrow (q \vee r)$$

$$\equiv \sim p \vee q \vee r$$

$$\text{RHS: } (p \rightarrow q) \vee (p \rightarrow r)$$

$$\equiv \sim p \vee q \vee \sim p \vee r$$

$$\equiv \sim p \vee q \vee r$$

$\therefore \text{LHS} = \text{RHS}$, S_1 is valid.

$$S_2: p \rightarrow (q \wedge r) \equiv [(p \rightarrow q) \wedge (p \rightarrow r)]$$

$$\text{LHS: } p \rightarrow (p \wedge r)$$

$$\equiv \sim p \vee (q \wedge r)$$

$$\text{RHS: } (p \rightarrow q) \wedge (p \rightarrow r)$$

$$\equiv (\sim p \vee q) \wedge (\sim p \vee r)$$

$$\equiv \sim p \vee (q \wedge r)$$

$\therefore \text{LHS} \equiv \text{RHS}$, S_2 is valid.

Q5 Text Solution:
Option (A)

True by using 'constructive dilemma'

Option (B)

$$1. \neg R \rightarrow (S \rightarrow \neg T)$$

$$2. \neg R \vee W$$

$$3. \neg P \rightarrow S$$

$$4. \neg W$$

We need to determine if $T \rightarrow P$ follows.

- From premise 4 ($\neg W$), and premise 2 ($\neg R \vee W$), we can conclude $\neg R$.

- Given $\neg R$, from premise 1, we get $S \rightarrow \neg T$.

- From premise 3, $\neg P \rightarrow S$, or equivalently, $\neg S \rightarrow P$.

Combining $S \rightarrow \neg T$ with $\neg S \rightarrow P$:

- If T is true, then S must be false because $S \rightarrow \neg T$.

- If S is false, then P must be true because $\neg S \rightarrow P$.

Thus, if T is true, P must be true, meaning $T \rightarrow P$.
 option (B) is true.

Option (c)

Given premises:

$$1. \neg P \wedge Q$$

$$2. Q \rightarrow (P \rightarrow R)$$

we need to determine if $\neg R$ follows:

- From premise 1, $\neg P$ and Q are both true

- Given Q is true, we consider the implication $Q \rightarrow (P \rightarrow R)$

- $P \rightarrow R$ must be true because Q is true

- Since $P \rightarrow R$ is equivalent to $\neg P \vee R$, and $\neg P$ is true, $\neg P \vee R$ is true, satisfying $P \rightarrow R$.

There is no information that guarantees R to be false or true independently from P . Hence, $\neg R$ does not necessarily follow from the premises



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given. **Option (C) is false.**

Option (d)

Given premises:

1. $P \rightarrow R$
2. P
3. $Q \vee \neg R$

We need to determine if Q follows.

- From premise 2, P is true.
- Given $P \rightarrow R$ from premise 1, R must be true.
- Now, consider $Q \vee \neg R$. Since R is true, $\neg R$ is false.
- Hence, $Q \vee \neg R$ simplifies to $Q \vee \text{false}$, which means Q must be true.

Thus, Q must be true. Option (D) is true.

Q6 Text Solution:

For I1,

Let p: "I will study discrete math"

q: "I will study English literature"

So premises will be,

1. $p \vee q$
2. $\neg p$

From premise 2, p is false. Hence, q has to be true from making premise 1 true.

So, I1 is incorrect.

For I2

Let p: "A works hard"

q: "B enjoys himself"

r: "C enjoys himself"

s: "D enjoys himself"

So premises will be,

1. $p \rightarrow (q \vee r)$
2. $q \rightarrow \neg p$
3. $r \rightarrow \neg s$

The inference is $p \rightarrow \neg s$

To proof by contradiction p and s has to true.

Premise 2 follows, if q is false,

Premise 1 follows, if r is false,

Now, in premise 3,

$$r \rightarrow \neg s \equiv T \rightarrow F \equiv F$$

So, antecedent becomes false and the conclusion follows.

Hence, I₂ is correct.

Q7 Text Solution:

$$\text{I. } \sim(p \rightarrow q) \equiv p \wedge \sim q$$

$$\text{II. } \sim(p \rightarrow q) \wedge (q \rightarrow r) \equiv (p \wedge \sim q) \wedge (\sim q \vee r) \\ \equiv (p \wedge \sim q)$$

$$\text{III. } p \wedge \sim q$$

$$\text{IV. } (p \vee q) \rightarrow r \equiv (\sim p \wedge \sim q) \vee r$$

So, I, II and III are equivalent to each other.

Q8 Text Solution:

$$S_1: \sim(p \leftrightarrow q) \equiv p \oplus q$$

$$S_2: p \leftrightarrow (\sim q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q) = p \oplus q$$

Q9 Text Solution:

$$S_1: \sim(p \vee (\sim p \wedge q))$$

$$\equiv \sim p \wedge \sim(\sim p \wedge q)$$

$$\equiv \sim p \wedge (p \vee \sim q)$$

$$\equiv \sim p \wedge \sim q$$

$$S_2: \sim p \wedge \sim q$$

Q10 Text Solution:

Implication is not associative.

P	Q	R	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$	$p \rightarrow q$	$(p \rightarrow q) \rightarrow r$	$(p \rightarrow (q \rightarrow r)) \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	T
T	F	T	T	T	F	T	T
T	F	F	T	T	F	F	F
F	T	T	T	T	T	T	T
F	T	F	F	T	T	F	F
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	F



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